

The Kodamai Container Program: A Complete Technical Record

Part V — Formalisation in Lean 4

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Abstract

Part V records the Lean 4 formalisation. The core equivalence-chain links and the cohomological obstruction engines are machine-checked: each oracle certifies a precise, honestly-scoped statement; most are Mathlib-free (Lean 4 core) and axiom-minimal (`propext/Quot.sound` only, several axiom-free via `decide`). Every item is **[lean-verified]** unless stated otherwise.

Thesis of Part V. The theorems are not hand-waving: the core equivalence-chain links and the cohomological obstruction engines are machine-checked in Lean 4. Each oracle certifies a precise, honestly-scoped statement; the general cohomology theory, where invoked, is cited as a black box (Brown VI.9), with the Lean file certifying only a finite/explicit computation.

1 The equivalence-chain milestones (M1–M4)

Precis. The container library and the directed-container $\leftrightarrow$ comonad $\leftrightarrow$ small-category equivalence, plus the ZS associativity characterisation.

Theorem 1.1 (M1/M2/M2b/M4 + ZS assoc, **[lean-verified]**). *Basic, Directed, ComonadConverse, Cofunctor, DContCat, ZappaSzep* — milestones M1/M2/M2b/M4 and ZS associativity (ZS1–ZS4 $\iff$ assoc).

Theorem 1.2 (M3, **[lean-verified]**). *Comonoid.lean / ComonoidConverse.lean* — directed container $\iff$ $\triangleleft$ -comonoid (both directions).

Theorem 1.3 (Cont and the four monoidal structures, **[lean-verified]**). *Cont.lean, Sequential.lean, Monoidal.lean, Dirichlet.lean, FourMonoidal.lean* — Cont as a category, the $\triangleleft$ operator, all four monoidal structures + comparison isos.

2 The free monad

Theorem 2.1 (Free monad, **[lean-verified]**). *The free monad on a container = a $\triangleleft$ -monoid; the three monoid laws (assoc = graft_assoc), Quot.sound-only. Node free-monad-grafting (Free.lean); a partial universal property in FreeUniversal.lean. (The cofree comonad is NOT core-formalisable — it needs coinductive PFunctor.M.)*

3 The cohomological obstruction engines (H^2)

Precis. Three standalone, Mathlib-free Lean files compute the group cohomology that homes the ZS and inventory obstructions. Honest scope: each certifies a FINITE/explicit computation; the general cohomology theory is cited as a black box (Brown VI.9).

Theorem 3.1 (Cyclic norm engine, [lean-verified]). $H^2(C_n; \mathbb{Z}) \cong \mathbb{Z}/n$ via the norm map. Node *cyclic-h2-norm-engine*.

Theorem 3.2 (Fibre $\mathbb{Z}/2$ engine, [lean-verified]). $H^2(\mathbb{Z}/2; \mathbb{Z}/2) \cong \mathbb{Z}/2$, the fibre edge, with nonsplit- $\mathbb{Z}/4$ vs. split-Klein witnesses. Node *fibre-h2-z2-engine*.

Theorem 3.3 (Unified gcd engine, [lean-verified]). $H^2(C_n; \mathbb{Z}/m) \cong \mathbb{Z}/\gcd(n, m)$, subsuming both engines above, with a from-scratch Bézout. Node *cyclic-h2-gcd-engine*.

4 The gerbe / reduced-nerve engine and o_ν slices

Precis. Machine-checked witnesses for the supply-chain gerbe and the skew-brace solvability obstruction.

Theorem 4.1 (Reduced-nerve $\mathbb{F}_2$ gerbe engine, [lean-verified]). $H^2(B(\mathbb{Z}/2)) = 1$, diamond $(1, 1, 0)$, D_4 non-split — the three gerbe witnesses (MEMORY lean-gerbe-h2-witnesses; grounds supply-chain-inventory-gerbe-h2).

Theorem 4.2 (Nerve-level separation oracle, [lean-verified]). $N_1 \subsetneq N_2$ machine-checked, bar complex $\mathbb{F}_2$, Betti $(1, 1, 1)$ vs. $(1, 0, 0)$ (grounds composition-nerve-level).

Theorem 4.3 (o_ν finite slices, [lean-verified]). The $|G| = 8$ additive slice, the $|G| = 16$ three-way lock + torsor dichotomy, and the corrected $D = \mathbb{Z}/8$ torsor criterion — all **decide**-based, axiom-free or **propext**-only (nodes under solvability-obstruction-o-nu / quiver-skew-brace-zs).

Theorem 4.4 (Prunability descent cochain, [lean-verified]). Prunability $\iff \delta \equiv 0$ on the $|L| = 2$ linearisation (node under quiver-skew-brace-zs).

5 Transfer and distributivity cells

Theorem 5.1 (Transfer, [lean-verified]). $\text{Monad} \rightarrow \text{comonad}$ transfer and its dual, three laws each (MonadComonadTransfer.lean, DualTransfer.lean).

Theorem 5.2 (Closed Dirichlet and distributivity cells, [lean-verified]). Dirichlet closed / Π -form internal hom (axiom-free); the four Hedges distributivity cells ($\triangleleft/\times$, $\triangleleft/+$, $\otimes/+$, $\times/+$); and the Workers retract $\Delta S \otimes \Delta T$ of $\Delta S \triangleleft \Delta T$.